



Fundamentals to Generative AI

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Credits: 5

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“Which of these use AI?”

1. Google Maps
2. Instagram filters
3. Face Unlock
4. Netflix recommendations
5. ChatGPT
6. Calculator
7. YouTube recommendations
8. Spam email detection
9. Google Translate
10. DALL-E

“Which of these can CREATE something new?”

“And that word — CREATE — is the key to today’s lecture.”

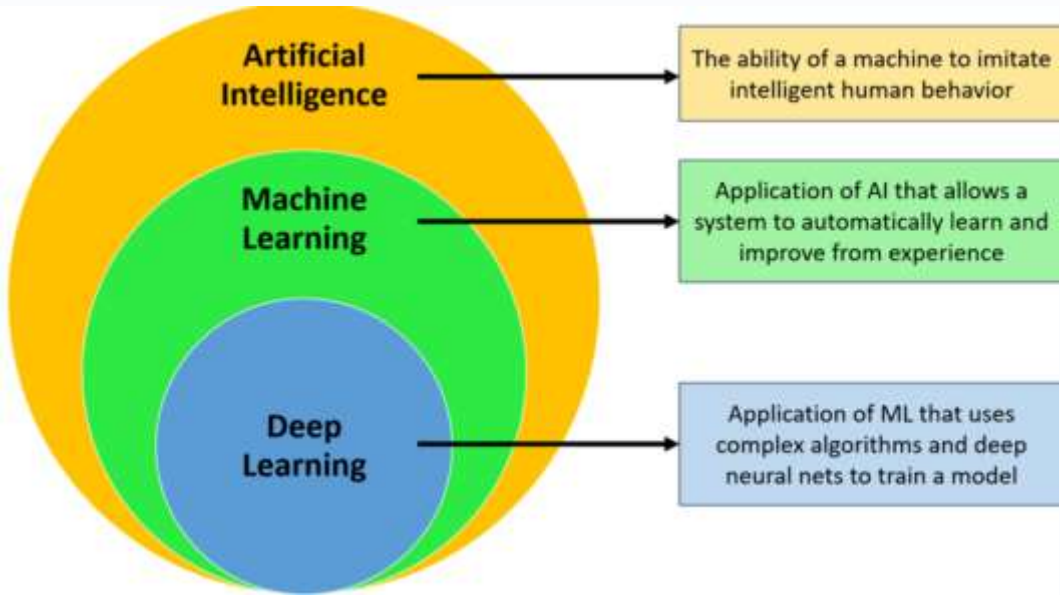
What is **Artificial Intelligence**?

Artificial Intelligence is the ability of a computer system to perform tasks that normally require human intelligence.

Examples: **Human ability → AI application**

“AI does not mean that the machine thinks exactly like a human.
"It means that the machine performs a task that appears intelligent."

AI, ML and DL

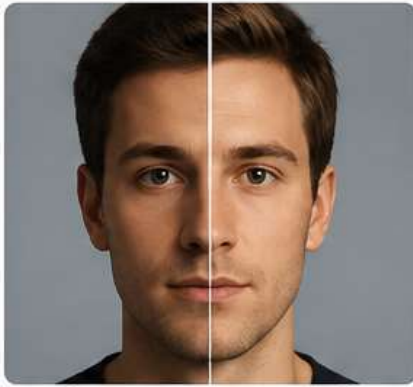


Human ability → AI application

Human ability	AI example
Recognize a face	Face recognition
Understand language	ChatGPT
Recognize speech	Siri/voice assistants
Make recommendations	Netflix/YouTube
Detect spam	Email spam filter
Understand images	Computer vision
Make decisions	Autonomous systems

“Don’t think of these as three completely separate technologies. ML is a subset of AI, and deep learning is a subset of ML.”

- **Generative AI**- where machines don't just analyze data—they create entirely new content.

Examples of Generative AI				
Text Generation  Once upon a time, in a futuristic city powered by clean energy, intelligent robots and humans worked together to create a better world full of possibilities. AI generates human-like text, stories, articles, etc.	Image Generation  AI creates images from text descriptions.	Code Generation <pre>def factorial(n): if n == 0: return 1 else: return n * factorial(n-1) # Example num = 5 print("Factorial of", num, "is", factorial(num))</pre> AI writes code, functions, or entire programs.	Music Generation  AI composes music, melodies, or soundtracks.	Video Generation  AI generates realistic videos from text or images.
Conversational AI  How can I help you today? Explain photosynthesis in simple words.  Photosynthesis is the process used by plants to make their own food using sunlight, water, and carbon dioxide. AI chatbots hold natural conversations and answer questions.	Art Generation  AI creates paintings and artistic styles.	3D Model Generation  AI generates 3D models for games, design, and architecture.	Voice Generation (Text-to-Speech)  AI converts text into natural-sounding human speech.	Deepfake Generation  AI creates realistic fake videos or alters existing ones.

The AI Hierarchy: From Intelligence to Generation



Artificial Intelligence (AI)

The science of building systems that can perform tasks requiring human intelligence like reasoning, learning, and decision-making.

Example: Google Maps suggesting the fastest route to your destination.



Machine Learning (ML)

A subset of AI where systems learn patterns from data instead of being explicitly programmed for every scenario.

Example: Netflix recommending movies based on your viewing history and preferences.



Deep Learning (DL)

A subset of ML using multi-layered neural networks for complex tasks like image recognition, speech processing, and natural language understanding.

Example: Face unlock feature on smartphones that recognizes your unique facial features.

Artificial Intelligence:

Example:

We want a computer to play chess intelligently.

There are different ways we could build it.

One traditional approach could be:

“If opponent does X, consider Y.”

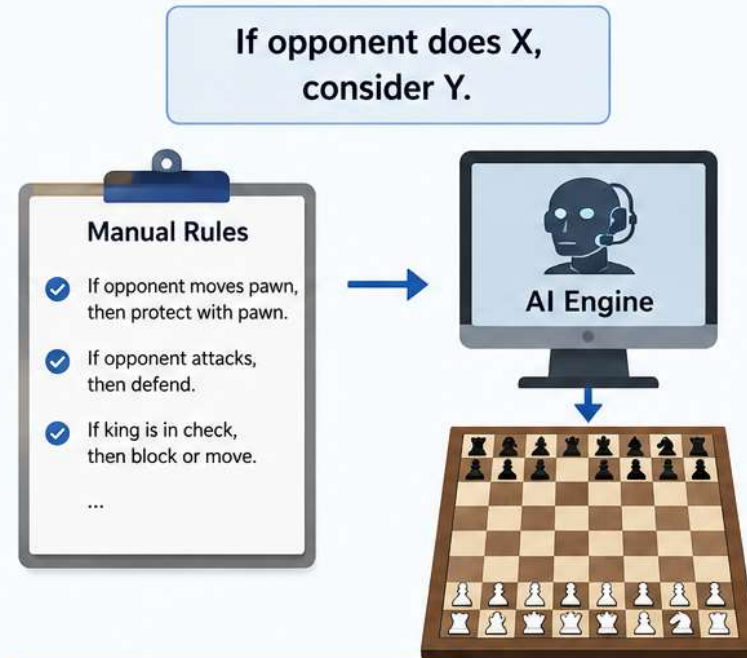
We can manually give rules.

That’s an AI approach.

AI Example: Teaching a Computer to Play Chess

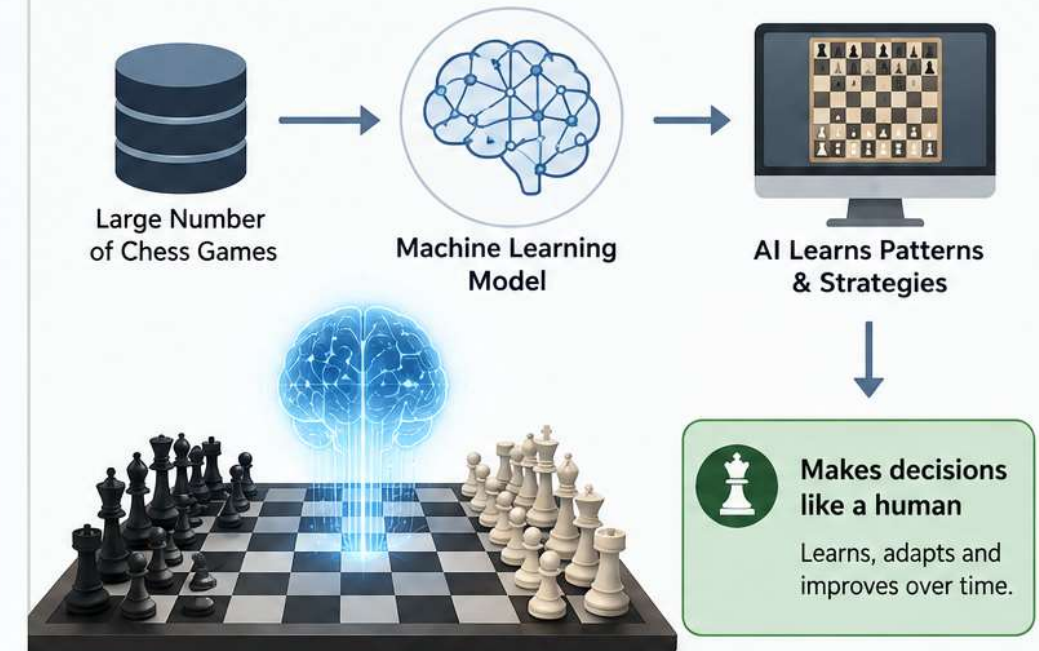
Goal: We want a computer to play chess intelligently.

1. Traditional AI Approach (Rule-Based)



 **AI follows the rules**
Good, but limited and hard to cover every situation.

2. Generative / Learning AI Approach



 **AI learns from experience**
Can discover new strategies and generalize to unseen positions.



Both are AI approaches: One uses explicit rules, the other learns from data. **Modern Generative AI learns and creates new strategies!**

Machine Learning:

“What if instead of manually writing every rule, we give the computer examples and allow it to learn patterns?”

That’s Machine Learning.

Example: **Spam** detection 
Spam

Suppose we give the computer:

"Congratulations! You won ₹10 lakh!" → **SPAM** 
Spam

"Meeting at 10 AM tomorrow" → NOT SPAM 
Email

"Click here to claim your prize" → **SPAM** 
Spam

"Your assignment has been submitted" → NOT SPAM



The model learns patterns.

Then a new email arrives:

“You have won a free iPhone.
Click now!”

The model predicts:

SPAM 
Spam

“We didn’t explicitly tell the machine every rule. We gave it examples, and it learned a pattern.”

That is the basic idea of ML.

Deep Learning:

“What happens when our data becomes extremely complex — images, speech, videos, natural language?”

We can use neural networks with many layers.

That’s Deep Learning.

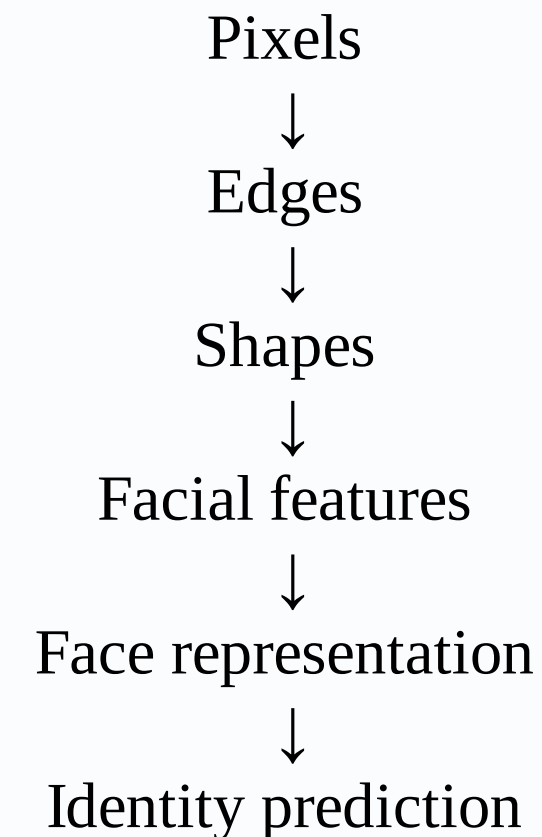
Example:

Face recognition

Input:

📷 Face image

The deep neural network can learn representations such as:



What is Generative AI?

Write:

Generative AI = AI that can generate new content.

Simple definition:

Generative AI refers to AI systems that learn patterns from existing data and use those learned patterns to generate new content.

The content can be:

- * Text
- * Images
- * Audio
- * Video
- * Music
- * Code
- * 3D content

Discriminative → DECIDE

Generative → CREATE

Example 1: Cats and Dogs

Suppose we have pictures of cats and dogs.

Discriminative model

Input:

🐱/🐶 Image

Output:

CAT



or DOG



It is learning to distinguish between categories.

Generative model

Input: "Generate a picture of a golden retriever playing in a park."

Discriminative vs Generative AI



	Discriminative AI	Generative AI
Main purpose	Distinguish / predict	Create
Question	"What is this?"	"Create something new"
Output	Label / prediction	New content
Example	Cat vs Dog	Generate a cat image
Spam detection	✓	✗
Face recognition	✓	✗
Recommendation	✓	✗
ChatGPT	✗	✓
DALL-E	✗	✓

AI



Machine Learning



Deep Learning



Discriminative



Generative



Predict/Classify



Create



Text/Image/Audio

Video/Code

Discriminative vs. Generative Models

Think of it in the **simplest possible way**:

Discriminative Model = “Which category does this belong to?”

Generative Model = “Can I create something new like this?”

A **discriminative model** learns how to **distinguish between different classes**

It looks at the input and decides **what it is**.

Example:

Task

Detect cancer in an X-ray

Identify a face

Detect spam

Predict whether a student passes

Image classification

Sentiment analysis

Output

Cancer / No Cancer

Person A / Person B

Spam / Not Spam

Pass / Fail

Cat / Dog / Bird

Positive / Negative

They focus on the **boundary between classes**.

Example

Detect cancer in an X-ray

Generate a medical report

Identify a face

Create a face that doesn't exist

Detect spam

Write an email

Predict tomorrow's temperature

Generate an image

Type

☐ **Discriminative**

☐ **Generative**

☐ **Discriminative**

☐ **Generative**

☐ **Discriminative**

☐ **Generative**

☐ **Usually discriminative/predictive**

☐ **Generative**

A **generative model** learns the **patterns and characteristics of the data** and can use them to **generate new data**.

For example, if a model learns thousands of cat pictures, it can generate a **new cat image that never existed before**.

Example 🐱

Training data:



The model learns:

- shape of ears
- eyes
- fur
- face structure
- colors
- textures

Then:

Prompt → “A cute cat wearing glasses” → **Generative Model** → 🐱👓

The generated cat is **new**.

Task	Output
Create an image	New image
Write an email	New text
Generate a medical report	New report
Create music	New music
Generate a face	New face
ChatGPT response	New text
Create code	New code

Discriminative vs. Generative Models

Understanding the fundamental difference between these two approaches is crucial for grasping how Generative AI works.

Discriminative Models

Learn boundaries between different classes or categories of data. They're excellent at classification and decision-making tasks.

Example: A spam filter that determines whether an email is spam or legitimate by learning the characteristics that distinguish between the two.

Generative Models

Learn the underlying distribution of data and can generate entirely new samples that resemble the training data.

Example: A model that can generate realistic fake email texts by understanding the patterns and structures of legitimate emails.

💡 **Think of it this way:** A discriminative model is like a judge deciding if food is Indian or Italian cuisine, while a generative model is like a chef who can actually cook authentic Indian or Italian dishes from scratch.

What Makes Generative AI Revolutionary?

Generative AI (GenAI)

Unlike traditional AI that simply analyzes existing data, Generative AI creates entirely new content—text, images, music, and code—that didn't exist before.

Examples:

- ChatGPT writing comprehensive essays
- DALL·E creating original artwork from text descriptions
- GitHub Copilot generating functional code

Why It Matters

Creativity Support

Design, music composition, and storytelling assistance

Productivity Boost

Automated email drafting and code generation

Scientific Research

Drug discovery through molecular structure generation



What is Information Theory?

Information Theory is the study of:

How information is measured, stored, and communicated.

In simple words:

Imagine I tell you:

“**The sun rose today.**” ☀

You are not surprised because it is almost certain.

“**It snowed in Pune today.**” ❄

More surprising event → More information

This idea leads us to **entropy**.

2. What is Entropy?

Simple definition

Entropy measures uncertainty or randomness.

Think of entropy as:

☐ “**How confused or uncertain am I about the outcome?**”

Example: Coin Toss ☐

Suppose I have a coin.

Case 1: Normal coin

You don't know whether the result will be:

Heads or Tails

There is uncertainty.

So:

→ **High entropy**

Case 2: A special coin

Suppose I tell you:

“This coin always gives Heads.”

Now you know the result.

There is **no uncertainty**.

So:

→ **Entropy = 0**

Remember:

More uncertainty → Higher entropy

Less uncertainty → Lower entropy

"The sky is"

The model may assign probabilities:

Next token	Probability
blue	60%
clear	15%
dark	10%
beautiful	5%
falling	2%
other	8%

There is a **strong likely answer:** "blue"

So uncertainty is relatively low.

Low entropy

The model is fairly confident about what comes next.

Now consider:

"The future of AI is"

Possible next words might be:

- exciting !!
- uncertain
- changing
- bright
- important
- unpredictable

Many possibilities have reasonable probabilities.

→ ☐ **More uncertainty**

→ ☐ **Higher entropy**

Entropy = Measure of Uncertainty

Low entropy → predictable

High entropy → unpredictable

Entropy = average amount of uncertainty/information

Entropy Formula

For a random variable X:

$$H(X) = - \sum_{i=1}^n P(x_i) \log_2 P(x_i)$$

Where X represents all x values possible to the variable

Where:

- $H(X)$ = Entropy
- $P(x_i)$ = Probability of each possible outcome
- $\log_2$ = logarithm with base 2
- n = number of possible outcomes

Suppose we toss a fair coin.

Possible outcomes:

$$P(H) = 0.5$$

$$P(T) = 0.5$$

Therefore:

$$H(X) = -[0.5 \log_2(0.5) + 0.5 \log_2(0.5)]$$

Since:

$$\log_2(0.5) = -1$$

$$H(X) = -[(0.5)(-1) + (0.5)(-1)]$$

$$H(X) = -[-0.5 - 0.5]$$

$$H(X) = 1 \text{ bit}$$

The unit when using $\log_2$ is **bits**.

$$H(X) = 1 \text{ bit}$$

Entropy = 1 bit

Interpretation

There is maximum uncertainty because both outcomes are equally likely.

What if the coin is NOT fair?

Suppose:

- Heads = 0.9
- Tails = 0.1

$$H(X) = -[0.9 \log_2(0.9) + 0.1 \log_2(0.1)]$$

$$\log_2(0.9) \approx -0.152$$

$$\log_2(0.1) \approx -3.322$$

$$\begin{aligned} H(X) &= -[(0.9)(-0.152) + (0.1)(-3.322)] \\ &= -[-0.1368 - 0.3322] \end{aligned}$$

$$H(X) \approx 0.469 \text{ bits}$$

Entropy is lower than 1 bit.

Compare

Coin	Uncertainty	Entropy
Always Heads	None	0
90% Heads, 10% Tails	Low	0.469
50% Heads, 50% Tails	High	1

"How much does knowing one variable tell us about another variable?"

Mutual Information measures how much two variables are related or how much information they share.

Suppose we observe two things:

•X = Is it raining?

•Y = Is the person carrying an umbrella?

Usually:

Raining → Person likely carries umbrella

If I tell you:

"The person is carrying an umbrella."

You can make a better guess that:

"It is probably raining."

Therefore, the two variables contain information about each other.

Suppose we have:

X = Student studies

Y = Student passes

Student	Studies?	Passes?
A	Yes	Yes
B	Yes	Yes
C	NO	NO
D	No	No

Mutual Information is high.

Mutual Information Formula

$$I(X; Y) = \sum_{x,y} P(x, y) \log_2 \left(\frac{P(x, y)}{P(x)P(y)} \right)$$

Where:

- X = first variable
- Y = second variable
- P(x,y) = joint probability
- P(x) = probability of X
- P(y) = probability of Y

Mutual information can also be expressed using entropy:

$$I(X; Y) = H(X) - H(X|Y)$$

Mutual Information = Original uncertainty – Remaining uncertainty after knowing Y

For example:

Suppose your uncertainty about whether a student will pass is:

$$H(X) = 1 \text{ bit}$$

After knowing whether the student studied:

$$H(X|Y) = 0.2 \text{ bits}$$

Then:

$$I(X; Y) = 1 - 0.2$$

$$I(X; Y) = 0.8 \text{ bits}$$

Interpretation:

Knowing whether the student studied reduced your uncertainty by **0.8 bits**.

Let's use a **fair coin** example.

Let:

- X = Coin 1

- Y = Coin 2

Suppose the two coins are completely independent.

Coin 1:

- H = 0.5

- T = 0.5

Coin 2:

- H = 0.5

- T = 0.5

Knowing Coin 1's result tells us **nothing** about Coin 2.

Therefore:

Because they don't share information.

$$I(X; Y) = 0$$

X

Studying

Day of week

Roll number

Y

Exam performance

Favorite color

Favorite food

Laptop brand ↔ Programming skill



Information Theory Fundamentals

Information theory provides the mathematical foundation for understanding how GenAI models process and generate information.

Entropy

A measure of uncertainty or randomness in data. Higher entropy means more unpredictability.

Example: Tossing a fair coin has high entropy (50-50 chance), while tossing a two-headed coin has zero entropy (100% predictable).

Mutual Information

Measures how much knowing one variable reduces uncertainty about another variable.

Example: Weather forecasts and umbrella sales have strong mutual information—knowing it will rain significantly reduces uncertainty about umbrella demand.

Advanced Inference Techniques

Variational Inference

A method to approximate complex probability distributions using simpler, more manageable ones. This makes computations feasible for real-world applications.

Example: Instead of mapping all possible ways a student might perform on an exam (infinitely complex), we approximate it with a bell curve distribution for practical analysis.

Amortized Inference

Reusing a trained model to make inference faster, rather than solving optimization problems from scratch each time.

Example: Instead of solving a complex math problem step-by-step every time, you memorize a formula (like a trained neural network) and apply it quickly when needed.

$$\mu = (\mu_i) = \sum_{j=1}^n \frac{1}{n} \mu_j = \frac{1}{n} \sum_{j=1}^n \mu_j$$

$$\mu_j = \frac{1}{n} \sum_{i=1}^n \mu_{ij} = \frac{1}{n} \sum_{i=1}^n \mu_{ij}$$

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Variational Inference

What is Inference? Inference means making a guess about something hidden using the information we can see.

Suppose you see this:

□ Wet roads + people carrying umbrellas, You don't directly see whether it rained.

But you infer: "It **probably rained.**"

So: Inference = **using available information to estimate something we cannot directly observe.**

What is Variational Inference?

Variational inference is a technique for approximating a difficult probability distribution with a simpler, easier-to-handle distribution.

This hidden information is called a **latent variable**

Imagine there is a hidden variable **Z** that we cannot directly observe.

Observed Data X



Model



Hidden variable Z

The problem: "Given the data X, what is the probability of the hidden variable Z?"

This can be mathematically difficult.

Variational inference says: **Instead of calculating the exact complicated distribution, find a simpler distribution that is close to it.**

Think of it as: "I cannot find the exact answer, so I'll find a good approximation."

What's Behind the Curtain?



We cannot see what is behind the curtain.
We use clues to make a **probability** (educated guess).

Clues (What we observe)



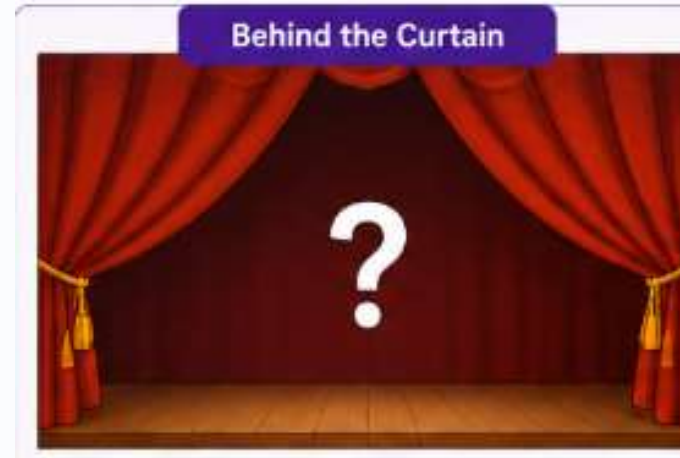
Clue 1:
It has four legs.



Clue 2:
It makes a barking sound.



Clue 3:
It wags its tail.



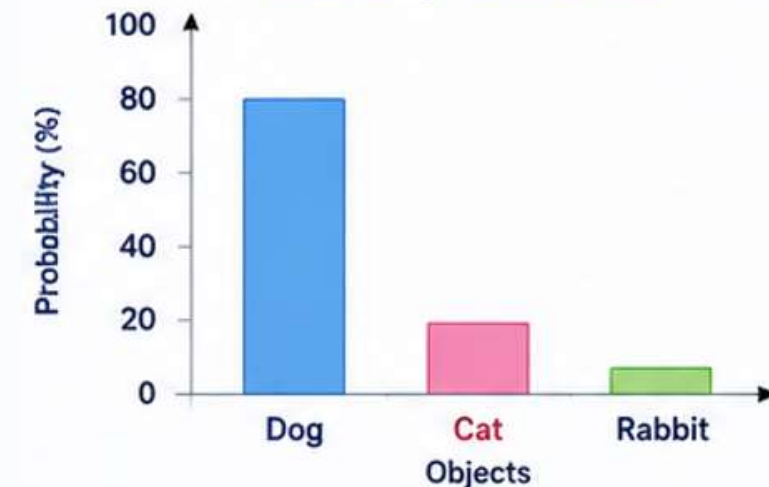
Based on the clues,
what is behind the curtain?

Assign probabilities.



Object		Assigned Probability
	Dog	80%
	Cat	15%
	Rabbit	5%

Visualizing Probabilities



Hidden object can be one of:



DOG



CAT



RABBIT

What does this mean?



We don't know
the exact answer.



We use clues
to make a belief.



That belief is a
probability distribution.



This is called
Inference!

★ Key Takeaway



- Variational Inference tries to learn a **good approximation** (probability distribution) of the true hidden reality.
- Instead of saying “It is a Dog” with 100% certainty, we say “It is **probably** a Dog with 80% chance”.
- This idea is used in many AI models like VAEs, topic models, Bayesian models, and more.



Real World Connection



Detective solves a case using clues (not full information).



Weather forecast gives probabilities (e.g., 70% rain).



Doctor diagnoses disease with probable causes.



Recommender systems guess what you may like (probabilistically).

Exact inference → difficult

Approximate inference → easier

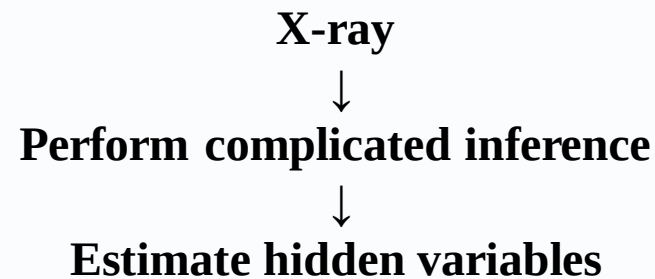
Variational inference → find the best approximation

Amortized Inference

Instead of solving the inference problem from scratch every time, train a neural network to learn how to do the inference.

Suppose an AI system needs to analyze **chest X-rays**.

For every X-ray, we want to estimate hidden information such as:



- important visual features
- possible disease-related patterns
- latent representation of the image

Without amortized inference

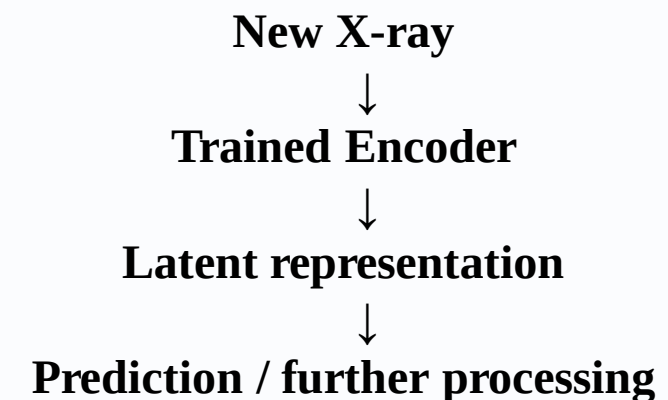
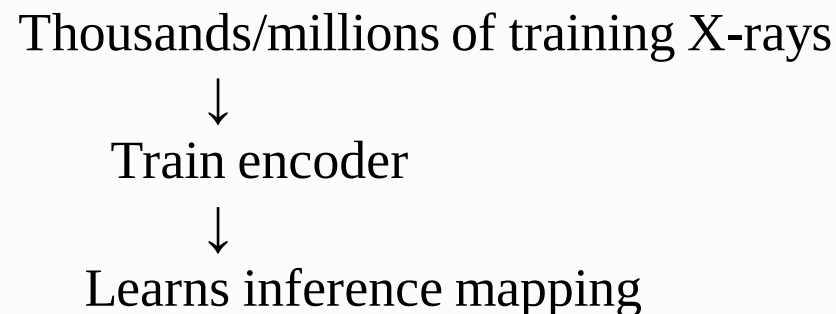
For every X-ray:

The trained encoder can quickly estimate the hidden representation for new X-rays. It has **learned a mapping from observed data to an approximate posterior/latent representation** during training.

If we have:

1 million X-rays

we repeat the expensive inference process 1 million times.



Evaluating Generative Model Performance

How do we measure if a generative model is actually good? Here are the key evaluation methods:



Log-likelihood

Measures the probability the model assigns to real data. Higher likelihood means better performance.

Example: A language model should assign higher probability to "I love AI" than to "AI love I."



FID (Fréchet Inception Distance)

Measures how close generated images are to real ones by comparing their feature distributions.

Example: Checking how realistic AI-generated cat images look compared to actual cat photos.



IS (Inception Score)

Measures both diversity and quality of generated samples simultaneously.

Example: A good model generates dogs of different breeds and poses, not just identical copies.



PRD (Precision-Recall)

Balances quality (precision) and variety (recall) in generated distributions.



Human Evaluation

The gold standard—asking people to judge if outputs are realistic and useful.

Example: The famous Turing test for evaluating text generation quality.

Log-Likelihood

How well does the model consider the observed data to be probable?

Suppose the real sentence is:

"The cat is sleeping."

Model A gives this sentence a **high** probability.

Model B gives it a very **low** probability.

Model A has a better likelihood for that example.

Log-likelihood

Instead of multiplying many probabilities, we use logarithms.

$$\log P(x)$$

For a dataset:

$$\log P(X) = \sum_i \log P(x_i)$$

Simple interpretation:

Higher log-likelihood generally means the model assigns higher probability to the observed data.

1. Evaluation of Generative Models

When a generative AI creates something (text, image, music), **we need to check: Is it good or bad?**

That's where evaluation metrics come in.

(a) Log-Likelihood

- **Idea:** How well does the model assign probability to real data?
- If the model gives **high probability** to correct sentences or images → it's good.
- If it gives **low probability** → not good.

Example:

Sentence: "I love ice cream."

- Good model → thinks this is highly likely.

Sentence: "Ice cream love I."

- Bad model → but if it thinks this is also likely, then it's failing.



(b) FID (Fréchet Inception Distance)

- Used for **images**.
- Compares generated images with real ones based on their learned feature representations.
- Lower FID = more realistic. (Lower FID $\rightarrow$ generated images are closer to real-image distribution.)

Example:

- You ask AI to generate dog images.
- If FID is low $\rightarrow$ the dogs look natural (different breeds, sizes).
- If FID is high $\rightarrow$ dogs look weird or fake (3 ears, no tail).



Are the generated images recognizable and diverse?

Higher **IS** is generally better.

(c) IS (Inception Score)

- Measures **quality + diversity** of generated images.
- Good model = images are clear **and** varied.

Example:

- If AI makes 100 images of dogs:
 - All same-looking dogs → not diverse.
 - Dogs in different poses, colors, breeds → good diversity → higher IS.

A good generated image should:

1. Clearly belong to a category.
2. The collection should contain different categories.

Collection A



All almost identical.

Collection B



Ask: "Which collection is more diverse?"



(d) PRD (Precision-Recall for Distributions)

- Balances **precision** (realistic outputs) and **recall** (variety of outputs).

Example:

- Precision = making a few very realistic dogs.
- Recall = making many dogs of all breeds, even if some look less real.
- PRD ensures both are considered.

Example

Suppose you want an AI model to generate animals.

Real dataset contains:

- 🐶 Dogs
- 🐱 Cats
- 🐎 Horses
- 🐘 Elephants

Model A generates:

- 🐶 Dogs only, but very realistic.

Precision → High

Recall → Low

Why?

It creates good samples but misses cats, horses, elephants.

Model B generates:

Dogs, cats, horses, elephants—but many are poor quality.

Recall → High

Precision → Low

Easy memory trick:

Precision = Quality

Recall = Coverage

(e) Human Evaluation

- Sometimes metrics don't capture human judgment → we ask people!
- Humans check if text makes sense, or if an image looks natural.

Example:

- AI writes a story.
- Automatic metrics may say it's okay, but humans may find it boring → so human evaluation is very important.

Suppose you ask two Generative AI models:
"Write a birthday message for my best friend."

AI Model A

"Happy Birthday! Wishing you a wonderful day filled with happiness, laughter, and success. Have a great year ahead!"

AI Model B

"Happy birthday to the person who has survived my nonsense for years! 😊 From random plans to unforgettable memories, life is definitely more fun with you. Here's to another crazy year! 🎉"

Which one is better? ☐

There is **no single mathematical answer**.
Different students may choose differently.

Model A = 3.6/5
Model B = 4.7/5

You can say:
"This is human evaluation."
Humans are judging the quality of AI-generated content using predefined criteria.

Criteria	Model A	Model B
Relevance	★ ★ ★ ★	★ ★ ★ ★ ★
Creativity	★ ★ ★	★ ★ ★ ★ ★
Naturalness	★ ★ ★ ★	★ ★ ★ ★ ★
Emotional feeling	★ ★ ★	★ ★ ★ ★ ★
Overall quality	★ ★ ★ ★	★ ★ ★ ★ ★



Core Components of Generative AI



Neural Networks

Computational models inspired by the human brain, designed for pattern recognition and learning.

Example: Instagram automatically tagging friends in your photos using facial recognition networks.



Embeddings

Converting data (words, images, sounds) into numerical vectors that capture semantic meaning.

Example: The famous word relationship "King - Man + Woman $\approx$ Queen" in vector space.



Representation Learning

Automatically discovering useful features and patterns from raw data without manual programming.

Example: Instead of manually coding rules for "two eyes, nose, mouth" to detect faces, the network learns these patterns automatically.

1. Neural Networks

A neural network is a computational model made of interconnected layers that learn patterns from data.



2. Embeddings

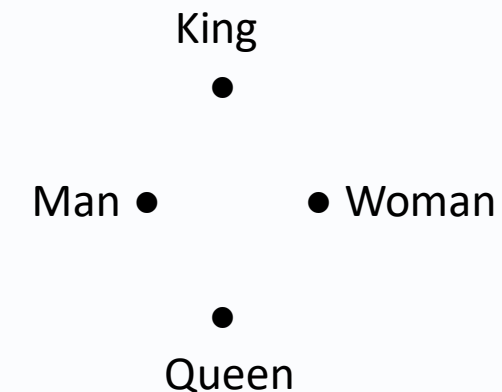
An embedding converts information such as words, images, or other objects into numerical vectors that capture useful relationships.

Computer doesn't naturally understand:

"king "as a concept.

It represents it as numbers:

king $\rightarrow$ [0.21, -0.45, 0.78, ...]



3. Representation Learning

Simple meaning:

Representation learning means allowing a model to automatically learn useful features/representations from raw data.

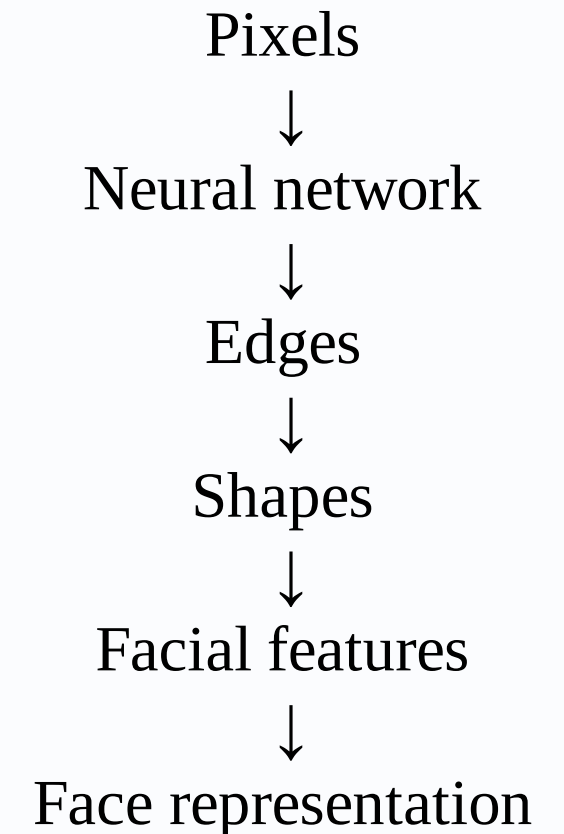
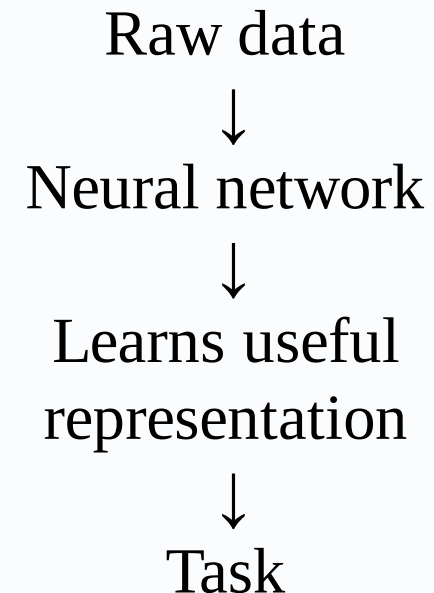
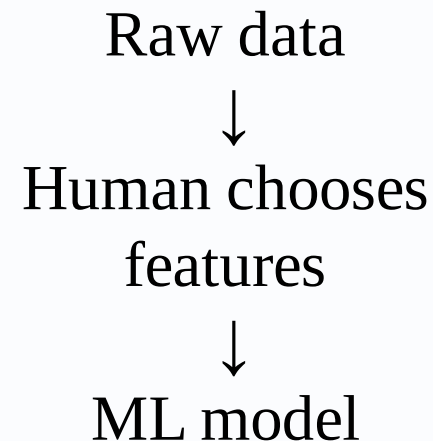
Traditional approach:

Representation learning:

Example

For a face:

Raw image:



The model learns useful representations automatically.

DATA-Generative AI depends heavily on data.

Examples:

Text generation

Data:

- books
- websites
- articles
- documents
- code

Image generation

Data:

- photographs
- artwork
- illustrations

Music generation

Data:

- audio
- musical patterns

Video generation

Data:

- video frames
- captions
- motion patterns

Tokenization

This is extremely important for language models.

Tokenization breaks text into smaller units called tokens so that the model can process it.

Example:

"I love AI."

Could become:

"I"

"love"

"AI"

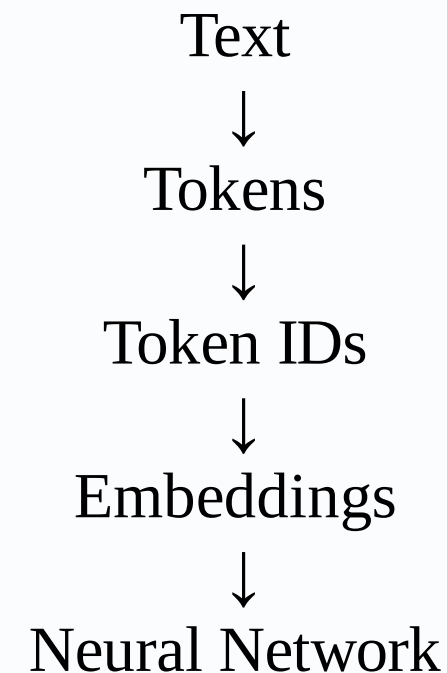
". "

Depending on the tokenizer, tokens may be **words, subwords, punctuation, or other pieces.**

Why tokenization?

Neural networks operate on numbers.

So:



From Raw Data to Generated Content

Understanding the pipeline from input data to AI-generated output is crucial for working with GenAI systems.

01

Data Collection

The foundation of GenAI—gathering text, images, audio, or other content types that serve as training material.

03

Embeddings

Converting tokens into dense numerical vectors in high-dimensional space that capture semantic relationships.

02

Tokenization

Breaking down text into smaller, manageable pieces (words, subwords, or characters) that the model can process.

Example: "ChatGPT is great" becomes ["Chat", "G", "PT", "is", "great"]

04

Conditioning Signals

Additional information provided to guide the generation process toward desired outcomes.

Example: When asking a text-to-image model to generate "a cat wearing sunglasses," the prompt serves as the conditioning signal.

Conditioning Signals:

A conditioning signal gives the model additional information that guides what it should generate.

Think: "Generate **something**, **BUT** generate it according to **this instruction**."

1. Class Conditioning

Suppose an image generator is told:

Class = Cat

It should generate: 🐱

If: **Class = Dog**

It should generate: 🐶

The class label guides generation.

2. Text Conditioning

Prompt:

"A robot walking through Pune during monsoon."

The text guides the generation.

Text prompt



**Generative
model**



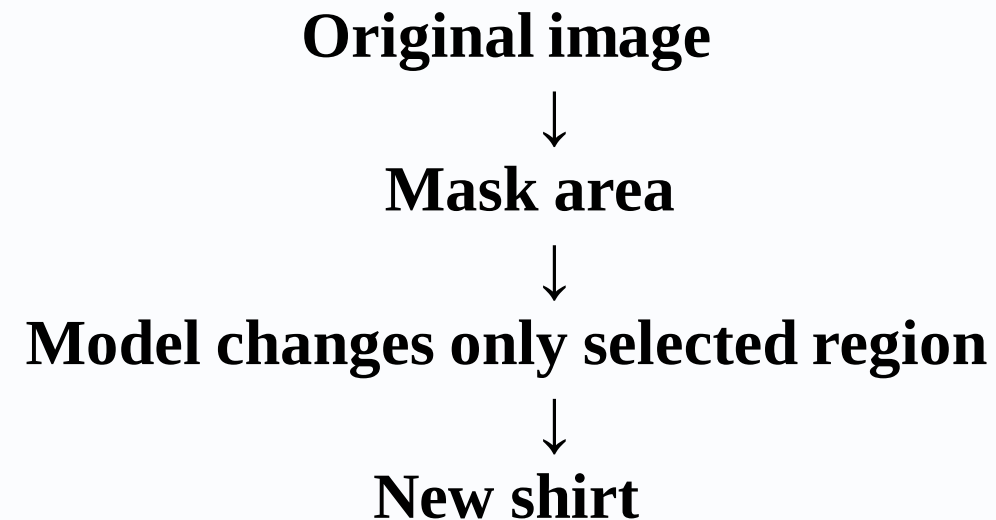
Image

3. Mask Conditioning : This is easiest to explain with image editing.

Suppose:

□ Image of a person wearing a red shirt.

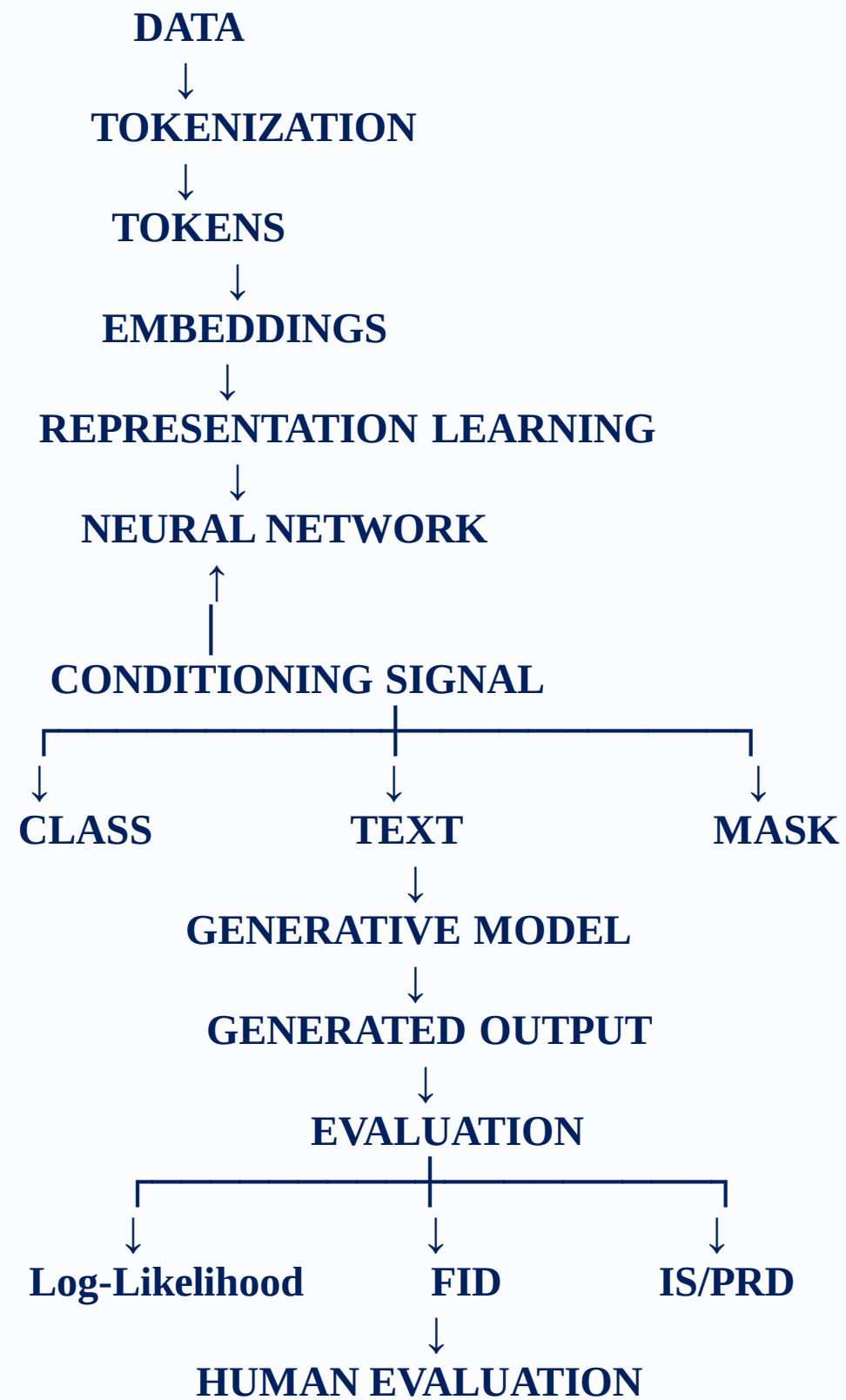
We mask the shirt area:



Example: "Change the red shirt to blue."

The mask tells the model:

"Modify this region."



"Design a GenAI System"

**AI Image Generator
AI Story Generator
AI Music Generator
AI Resume Generator
AI College Assistant**

They must answer:

- 1. What is your data?**
- 2. How will you tokenize/process it?**
- 3. What are the embeddings?**
- 4. What does the neural network learn?**
- 5. What conditioning signals will you use?**
- 6. How will you evaluate your model?**

Real-World Application: Google Translate

Let's see how all these concepts work together in a practical example that millions use daily.

